

Explain how the structure and properties of atoms relate to the organisation of the elements in the periodic table

Learning intention

- explain how the structure and properties of atoms relate to the organisation of the elements in the periodic table.

Teach and model

- investigating the physical properties of some metals and non-metals.
- conducting flame tests for a selection of elements and examining emission spectra.
- examining how the development of the spectroscope led to further development of the model of the atom.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.